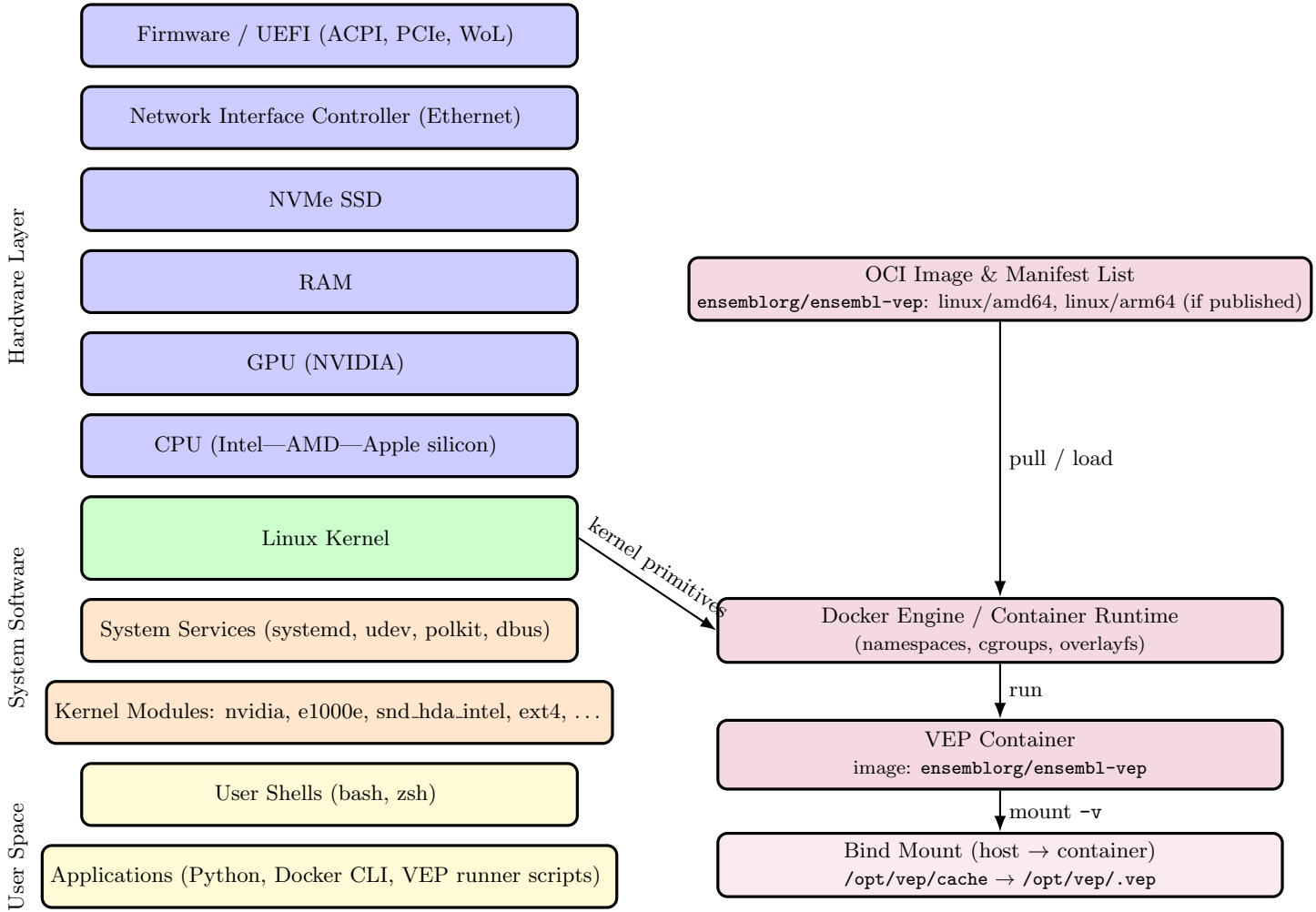

Hardware Layer
System Software
User Space
Firmware / UEFI (ACPI, PCIe, WoL)
Network Interface Controller (Ethernet)
NVMe SSD
RAM
GPU (NVIDIA)
CPU (Intel—AMD—Apple silicon)
Linux Kernel
System Services (systemd, udev, polkit, dbus)
Kernel Modules: nvidia, e1000e, snd_hda_intel, ext4, ...
User Shells (bash, zsh)
Applications (Python, Docker CLI, VEP runner scripts)
kernel primitives
OCI Image & Manifest List
ensemblorg/ensembl-vep: linux/amd64, linux/arm64 (if published)
pull / load
Docker Engine / Container Runtime
(namespaces, cgroups, overlayfs)
run
VEP Container
image: ensemblorg/ensembl-vep
mount -v
Bind Mount (host → container)
/opt/vep/cache → /opt/vep/.vep